Capstone Project: Web Scrapping Flipkart Site

PHASE 1:Extracting Data With Help of Web Scrapping

STEP 1: Import Required Libraries

In [1]:
```
import time
import requests
from requests.adapters import HTTPAdapter
from urllib3.util.retry import Retry
import time
from bs4 import BeautifulSoup
```

In [2]:
```
retry_strategy = Retry(total=3,read=3,connect=3, backoff_factor=3,status_forcelist=[500, 502, 503, 504], allowed_methods=["GET"])

adapter = HTTPAdapter(max_retries=retry_strategy)
session = requests.Session()
session.mount("https://", adapter)
session.mount("http://", adapter)

# Your scraping loop
for i in range(1, 251):
    url = f' https://www.flipkart.com/search?q=earbuds&otracker=search&otracker1=search&marketplace=FLIPKART&as-show=on&as=off&page={i}'
    headers = {'User-Agent':'Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36'}

    try:
        page = session.get(url, headers=headers, timeout=15)
        page.raise_for_status()

        with open(f"data/page{i}.html", 'w', encoding="utf-8") as f:
            f.write(page.text)
        print(f"Successfully page {i}")


        time.sleep(15)

    except requests.exceptions.Timeout as e:
        print(f"Timeout on page {i}: {e}")

    except requests.exceptions.RequestException as e:
        print(f" error on page {i}: {e}")
```

In [3]:
```
from bs4 import BeautifulSoup
```

In [4]:
```
with open(f"data/page1.html",encoding='utf-8', errors='ignore') as f:
    contents = f.read()
```

In [5]:
```
soup = BeautifulSoup(contents, "html.parser")
```

In [6]:
```
main=soup.select('div.RGLWAk')
#main
```

Step 2 : Sample data

In [7]:
```
import pandas as pd

# Initialize the list
all_rows = []

for div in main:
    # Use safe access pattern for each element
    name_element = div.select_one('a.pIpigb')
    name = name_element.text.strip() if name_element else "N/A"

    price_element = div.select_one('div.hZ3P6w')
    price = price_element.text.strip() if price_element else "N/A"

    original_price_element = div.select_one('div.kRYCnD')
    orignal_price = original_price_element.text.strip() if original_price_element else "N/A"

    # Fixed typo in variable name (orignal_price -> original_price)
    # Removed duplicate code block for original_price

    reviews_element = div.select_one('span.PvbNMB')
    # Fixed this line - was using original_price_element instead of reviews_element
    reviews = reviews_element.text.strip() if reviews_element else "N/A"

    rating_element = div.select_one('div.MKiFS6')
    Rating = rating_element.text.strip() if rating_element else "N/A"

    bank_offers_element = div.select_one('div.hx1EGN')
    Bank_offers = bank_offers_element.text.strip() if bank_offers_element else "N/A"

    row = {'name': name, 'price': price, 'original_price': orignal_price,'reviews':reviews, 'Rating':Rating,'Bank_offers':Bank_offers}
    all_rows.append(row)

# Create the final dataframe in one go
data = pd.DataFrame(all_rows)
```

In [8]:
```
data
```

Out[8]:

	name	price	original_price	reviews	Rating	Bank_offers
0	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	₹1,299	₹5,499	N/A	N/A	N/A
1	Caidea TW121 Pro Pods Upto 40 Hrs Playtime* Ty...	₹265	₹1,299	(923)	3.5	Super Deals
2	BeastBass CMT Buds 2 ENC, Ultra Bass Technolog...	₹327	₹2,999	(339)	4.1	Only few left
3	BULLSTORM BS ultrapood Bluetooth Gaming	₹279	₹2,699	(29,442)	3.4	Bank Offer
4	TECHIO Black Earbuds | True Wireless Stereo | ...	₹235	₹1,980	(11,657)	3.6	Only few left
5	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	₹1,299	₹5,499	N/A	N/A	N/A
6	myxes New Edition TWS M19 Gaming Earbuds Bluet...	₹299	₹1,999	(69,905)	3.6	Bank Offer
7	PARADISEECOM Max UltraPods Bluetooth Earbuds, ...	₹430	₹699	N/A	N/A	Only 2 left
8	Earboss 4 Mics Auto AI ENC , 120 HRS Playback ...	₹419	₹3,499	(2,571)	3.8	Bank Offer
9	MAGENESE M19 Earbuds/TWS/buds 5.1 with 280H Pl...	₹292	₹1,932	(925)	3.7	Only few left
10	Caidea TW500 Tube Upto 50H Playtime*, ENC, Fas...	₹300	₹2,999	(16,594)	3.7	Hot Deal
11	TRIGGR Kraken X1 with Battery Display, 40ms La...	₹699	₹3,999	(88,543)	4	Super Deals
12	boAt Airdopes Alpha,35H Battery, 13mm Drivers,...	₹799	₹3,490	(7,19,265)	4	Super Deals
13	TECHIO AirBeats Wireless Earbuds with Dual Mic...	₹572	₹4,999	(912)	3.7	Only few left
14	realme Buds T110 (RMA2306) with AI ENC for cal...	₹1,099	₹2,999	(4,39,861)	4.2	Hot Deal
15	BITWEAVE Airpox pro 2 with B205 Driver (USB-C)...	₹380	₹2,000	N/A	N/A	Only few left
16	GOBOULT W40 with 4 Mic ENC, 48H Battery Life, ...	₹799	₹2,999	(1,70,536)	4.1	Super Deals
17	Caidea TWS8 Sugar V5.4 Upto 30Hrs Playtime* Ty...	₹299	₹2,399	(923)	3.5	Super Deals
18	OnePlus Nord Buds 3r TWS Earbud, 10min charge ...	₹1,499	₹1,999	(3,62,241)	4.4	Hot Deal
19	TECHBROS New Edition TWS M19 PRO with Power Ba...	₹295	₹1,999	(1,091)	3.7	Only few left
20	GOBOULT Z40 with ENC Mic, 60Hr Battery, Metal ...	₹999	₹4,999	(4,83,289)	4.1	Bank Offer
21	Aroma NB121 Pods Upto 40 Hours Playtime* Type ...	₹268	₹1,299	(1,07,050)	3.6	Bank Offer
22	boAt Airdopes Alpha,35H Battery, 13mm Drivers,...	₹799	₹3,490	(7,19,265)	4	Bank Offer
23	myxes Tws Ecouteur Headsets Earphones Wireless...	₹299	₹1,999	(10,561)	3.7	Bank Offer
24	GWALBROS New White Bluetooth Earbuds, True Wir...	₹242	₹1,999	(4,353)	3.6	Only few left
25	PARADISEECOM (2nd generation) ANC Deep Bess Bl...	₹621	₹699	N/A	N/A	Only 2 left
26	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	₹1,299	₹5,499	N/A	N/A	N/A
27	Mivi Duopods B1 (Just Launched)45Hrs Playtime,...	₹599	₹2,999	(45,742)	4.1	Bank Offer
28	realme Buds T110 (RMA2306) with AI ENC for cal...	₹1,099	₹2,999	(4,39,861)	4.2	Hot Deal
29	myxes Earbuds Bluetooth V5.1 True Wireless ste...	₹299	₹1,999	(2,991)	3.6	Bank Offer
30	TRIGGR Ultrabuds N3 with Crisp Calling, 50H Pl...	₹699	₹1,999	(9,150)	4.1	Bank Offer
31	Number Super Buds GTS9 w/Monster Led Light,70H...	₹1,099	₹5,999	(46)	4.3	Bank Offer
32	truke Buds AURA with 70Hrs Playtime, 4-Mic ENC...	₹699	₹2,799	(4,832)	4	Bank Offer
33	Caidea TW400 Aristo 32dB Active Noise Cancella...	₹499	₹3,499	(16,594)	3.7	Only few left
34	NEXX ARHAM TWS BLUETOOTH HEADSET WIRELESS TWIN...	₹288	₹1,699	(14,333)	3.6	Only few left
35	Caidea TWS7 Sufi V5.4 Upto 30Hrs Playtime* Typ...	₹299	₹2,399	(923)	3.5	Super Deals
36	TRIGGR Kraken X4 with 13mm Drivers, 40ms Laten...	₹799	₹3,999	(48,668)	4	Bank Offer
37	GOBOULT Y1 Pro with 4 Mic ENC, 60Hrs Battery L...	₹999	₹5,999	(1,44,987)	4.1	Bank Offer
38	PTron Bassbuds Vibe w/ 34 Hrs Playback,Clear C...	₹599	₹2,899	(19,557)	4.1	Hot Deal
39	Mivi Duopods B1 (Just Launched)45Hrs Playtime,...	₹599	₹2,999	(45,742)	4.1	Bank Offer

STEP 3: Extract data into CSV file

In [9]:

```
all_data = []
# First, check which files actually exist
import os

data_dir = "data"
existing_files = []

# Find which page files actually exist
for i in range(1, 252):
    file_path = f"{data_dir}/page{i}.html"
    if os.path.exists(file_path):
        existing_files.append(i)

# Now only process files that exist
for i in existing_files:
    try:
        with open(f"{data_dir}/page{i}.html", encoding='utf-8', errors='ignore') as f:
            contents = f.read()

        soup = BeautifulSoup(contents, "html.parser")
        main = soup.select('div.RGLWAk')

        for div in main:
            # Use safe access pattern for each element
            name_element = div.select_one('a.pIpigb')
            name = name_element.text.strip() if name_element else "N/A"

            price_element = div.select_one('div.hZ3P6w')
            price = price_element.text.strip() if price_element else "N/A"

            original_price_element = div.select_one('div.kRYCnD')
            original_price = original_price_element.text.strip() if original_price_element else "N/A"

            reviews_element = div.select_one('span.PvbNMB')
            reviews = reviews_element.text.strip() if reviews_element else "N/A"

            rating_element = div.select_one('div.MKiFS6')
            Rating = rating_element.text.strip() if rating_element else "N/A"

            bank_offers_element = div.select_one('div.hx1EGN')
            Bank_offers = bank_offers_element.text.strip() if bank_offers_element else "N/A"

            # Fixed typo in variable name (orignal_price -> original_price)
            row = {'name': name, 'price': price, 'original_price': original_price, 'reviews': reviews, 'Rating': Rating, 'Bank_offers': Bank_offers}
            all_data.append(row)
    except FileNotFoundError:
        print(f"Warning: File data/page{i}.html not found, skipping.")
    except Exception as e:
        print(f"Error processing file data/page{i}.html: {e}")

# Create the final dataframe in one go
```

```
data = pd.DataFrame(all_data)
print("lalit is great")
```

```
lalit is great
```

In [11]:
```
data.to_csv("five_days_work.csv")
```